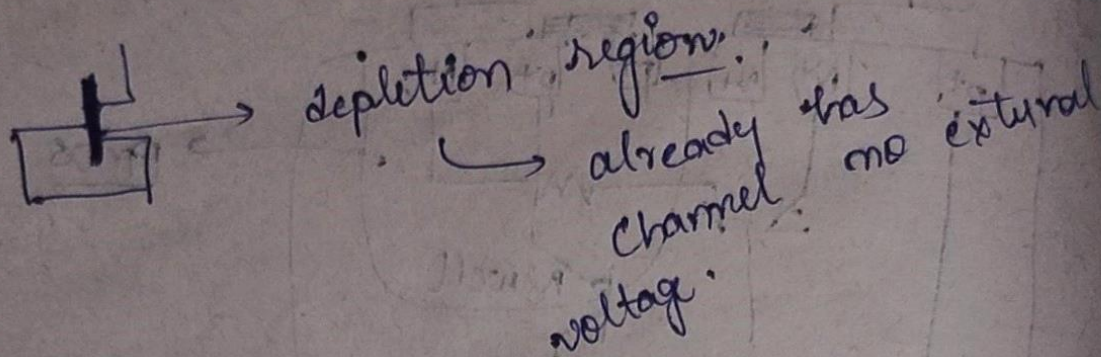


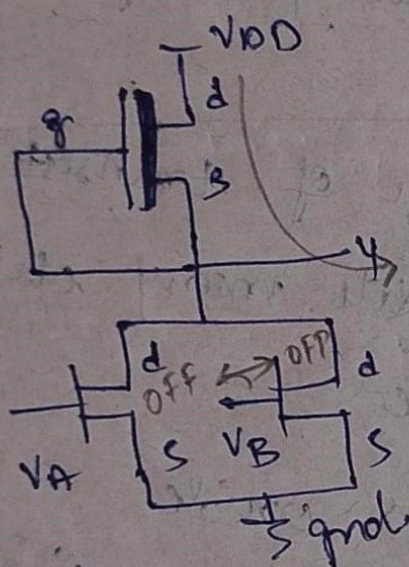
25/2/25

CMOS circuit design with depletion NMOS load

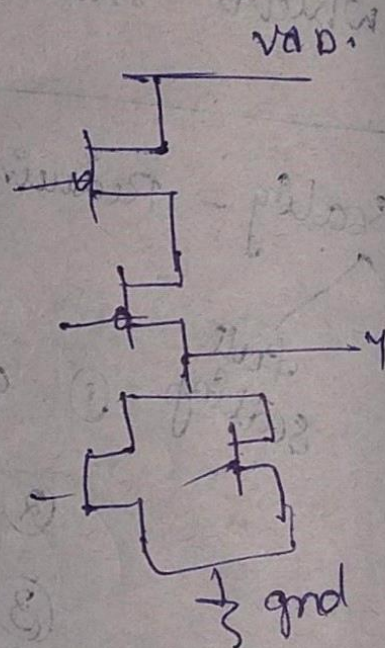
Depletion load.



NOR with depletion NMOS load.



NOR with CMOS



Truth Table for NOR gate:

VA	VB	Y
low	low	high
low	high	low
high	low	low
high	high	low

VOH = VDD
VOL = 0

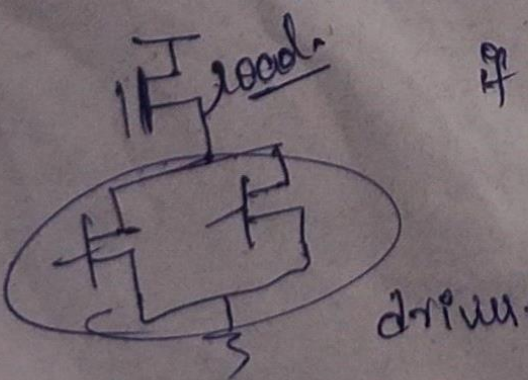
VOH - output low
VOL - op high

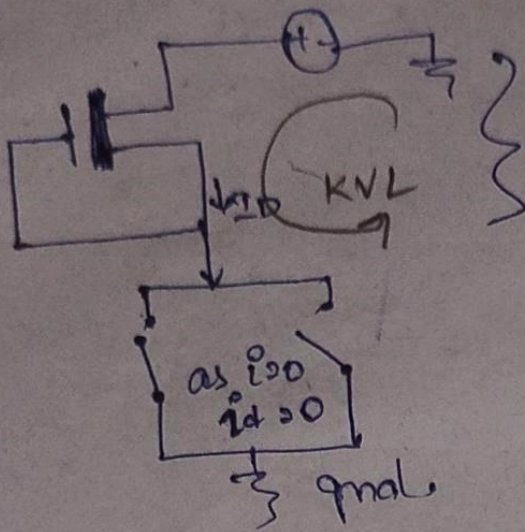
(DC) static Analysis.

VOH, VOL = ?

If low - VA, VB

VA & VB are open means $I = 0$.





linear region
of operation.

as $V_G = \text{qnd. voltage}$ is $\uparrow$ I_D
also increases.

I_D in linear,

$$I_D = \mu C_{ox} \left(\frac{W}{L} \right) \left((V_{GS} - V_{th}) \left(V_{DS} - \frac{V_{DS}}{2} \right) \right)$$

K_{nload} process const.
nmos.

$$I_D = \frac{K_{nload}}{2} \left(2(V_{GS} - V_{th}) V_{DS} - V_{DS}^2 \right) = 0$$

common as 0, $V_{GS} = 0$ (shorted)
 V_{DS} on KVL,

$$-V_{DD} + V_{DS} + V_{out} = 0$$

as $V_{OH} > V_{out}$

$$V_{DS} = V_{DD} - V_{OH}$$

$$I_D = \frac{K_{nload}}{2} \left(2V_{th}(V_{DD} - V_{OH}) - (V_{DD} - V_{OH})^2 \right) = 0$$

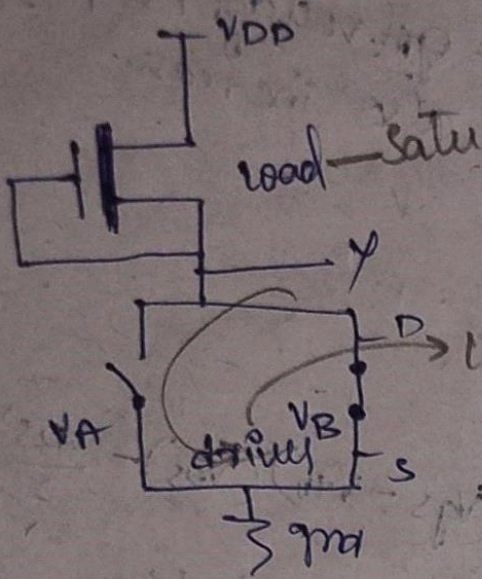
$$2V_{th}(V_{DD} - V_{OH}) - (V_{DD} - V_{OH})^2 = 0$$

$$2V_{th}V_{DD} - 2V_{th}V_{OH} - V_{DD} + V_{OH} = 0$$

Consider, as $V_{th} = 0$ as $V_{th} = 0$ depletes V_{OH}
 $V_{OH}(1 - 2V_{th}) = V_{DD} - 2V_{th}V_{DD}$
 $V_{OH} = \frac{V_{DD} - 2V_{th}V_{DD}}{1 - 2V_{th}}$

VOL = high high

anyone high



$$I_{load} = I_{P, driver} + I_{N, driver}$$

linear (discharge)

$$Y = 0 = V_B = 0$$

$$Y = V_B$$

I_{Drain} for sat,

$$I_D = \frac{\mu C_{ox} \left(\frac{W}{L} \right)}{2} \left(\frac{V_{gs} - V_{th}}{2} \right)^2$$

$$I_{D, linear} = \mu C_{ox} \left(\frac{W}{L} \right) \left((V_{gs} - V_{th}) V_{ds} - \frac{V_{ds}^2}{2} \right)$$

$$I_{D, load} = \frac{K_{n, load}}{2} \left(\frac{V_{gs} - V_{th}}{2} \right)^2$$

also depends on body source

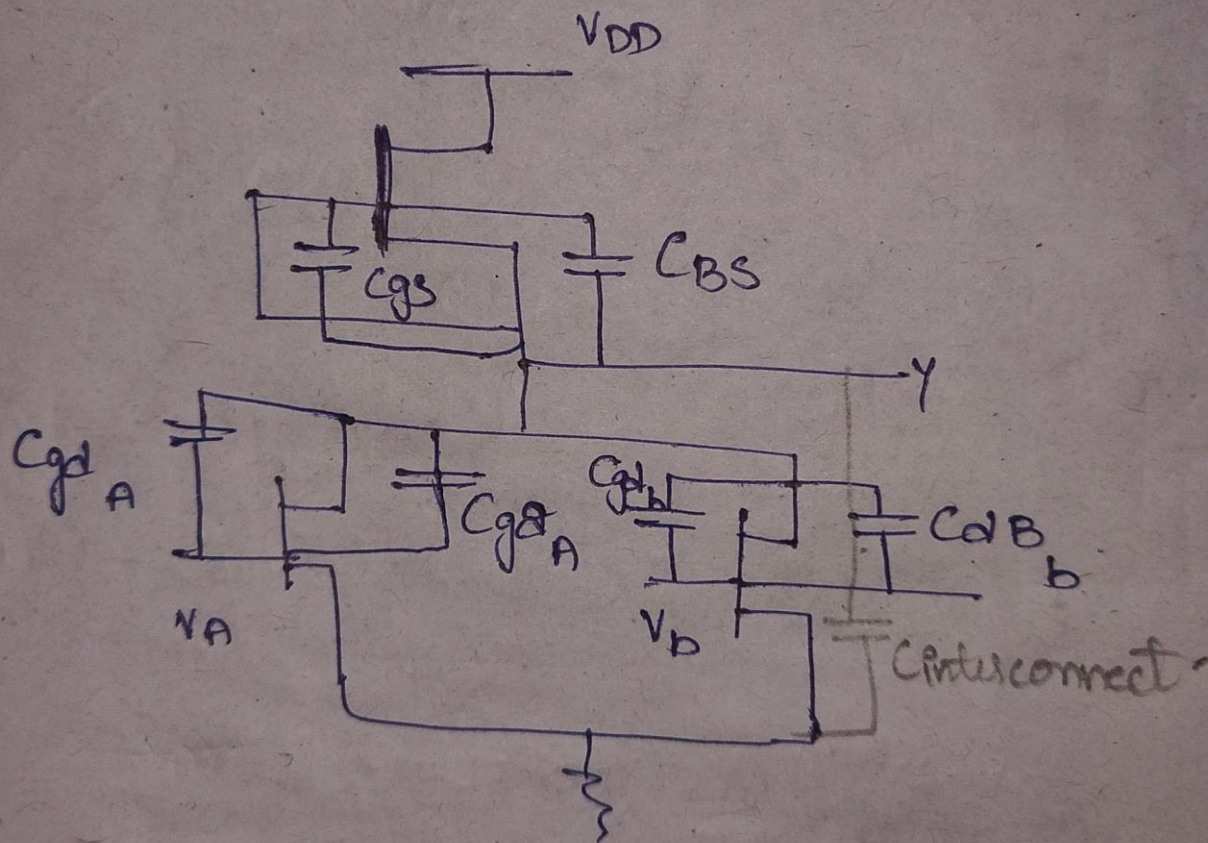
this is equal to

$$I_{D, driver} = \frac{K_{n, driver}}{2} \left(2(V_{gs} - V_{th}) V_{ds} - V_{ds}^2 \right)$$

$$\Rightarrow I_{D, load} = \frac{K_{n, load}}{2} \left(0 - V_{th, load} (VOL) \right) \left[V_{th} (VOL) \right]$$

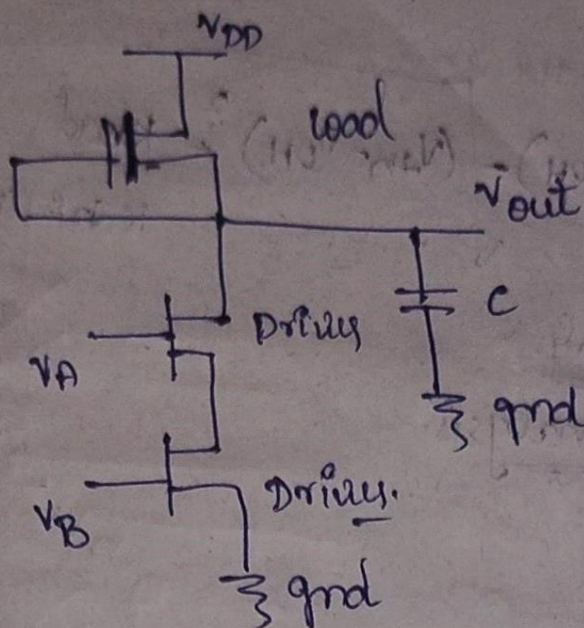
$$I_{D, load} = \frac{K_{n, load}}{2} \left(-V_{th, load} (VOL) \right) \left(-V_{th} (VOL) \right)$$

Dynamic capacitance
NOR gate



28/2/25

Nand gate with nmos depletion load.



V_A	V_B	Y (V_{OH})
Low	Low	High
L	H	H (V_{OH})
H	L	H (V_{OH})
H	H	L (V_{OL})

$H = V_A, V_B \rightarrow$ satce - load
linear - driver

$$I_{load} = I_{Driver A} = I_{Driver B}$$

$$I_{load} = \frac{I_{driver A} + I_{driver B}}{2}$$

$$= \frac{k_n}{2} (V_{gs} - V_{th})^2 \rightarrow \text{sat}$$

$$V_{gs} = 0 \text{ (depletion)}$$

$$= \frac{k_n}{2} (0 - V_{th}(V_{OL}))^2$$

$$I_{load} = \frac{k_n}{2} (V_{th}(V_{OL}))^2$$

$$V_{OL} = V_{OLA} + V_{OLB}$$

where,

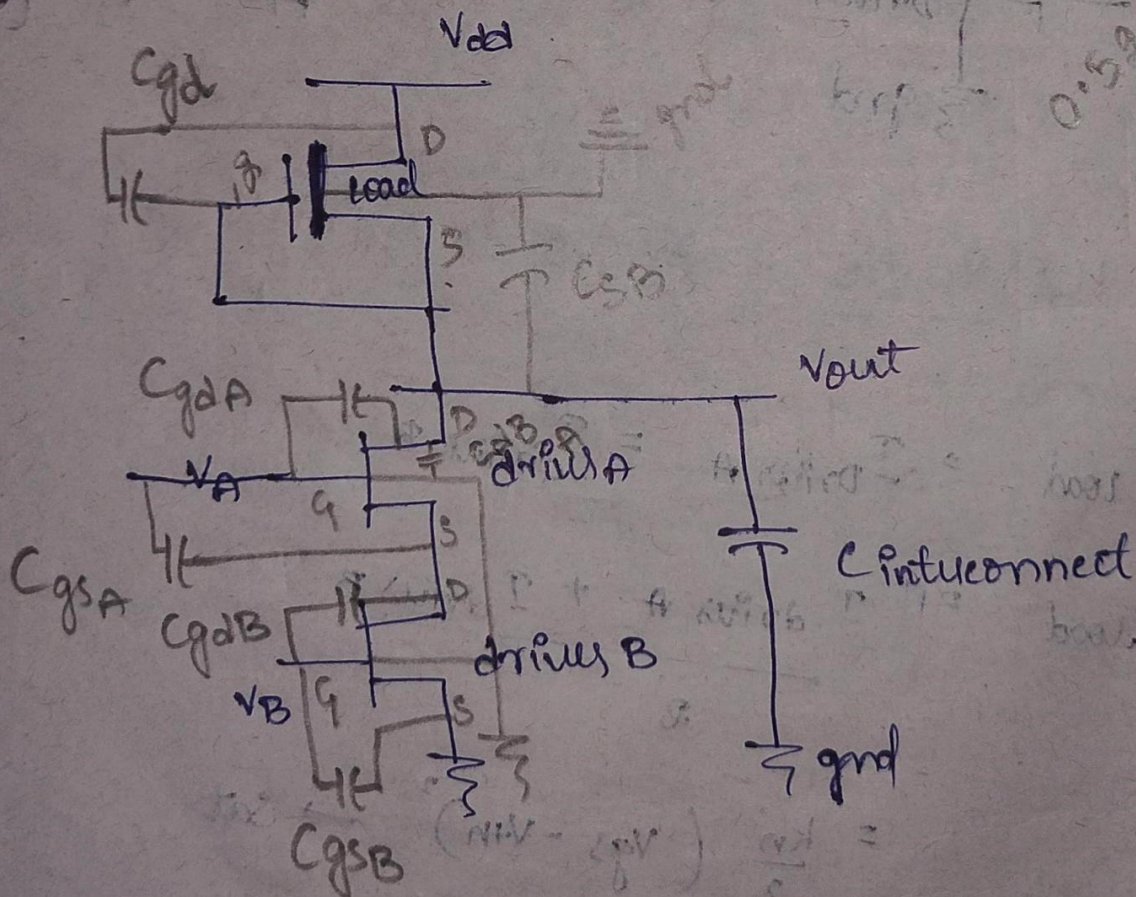
$$V_{OL A} = (V_{th} - V_{OH}) - \sqrt{(V_{OH} - V_{th})^2 - \frac{K_{load}}{K_{driver A}} V_{th} (V_{OL})^2}$$

$$V_{OLB} = (V_{th} - V_{OH}) - \sqrt{(V_{th} - V_{OH})^2 - \frac{K_{load}}{K_{drives B}} (V_{th} (V_{OL}))}$$

So,

$$V_{OL} = 2 \left[(V_{th} - V_{OH}) - \sqrt{(V_{th} - V_{OH})^2 - \frac{K_{load}}{K_{drives}} V_{th} (V_{OL})^2} \right]$$

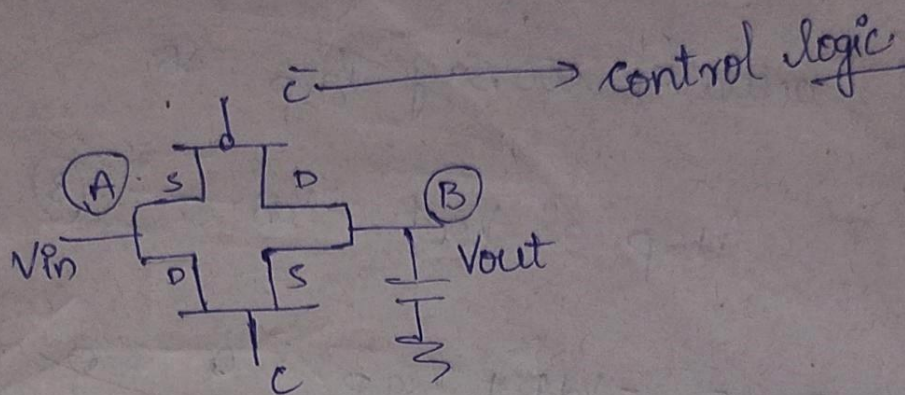
Dynamic Analysis $\rightarrow$ Delay.



4/8/25

CMOS Transmission gates (pass gate).

Transmission gate - PMOS, NMOS parallel.



operation, $c = \text{high} = V_{dd}$

then P, NMOS are on, with I_{DSDP} &

I_{DSN} .

but when $c = \text{low}$ both NMOS, PMOS are off.
[open circuit]

PMOS current is due to holes so
 $s \rightarrow d$
NMOS current is due to e^- so
 $d \rightarrow s$. [OPP]

ip & op current & voltage relation.

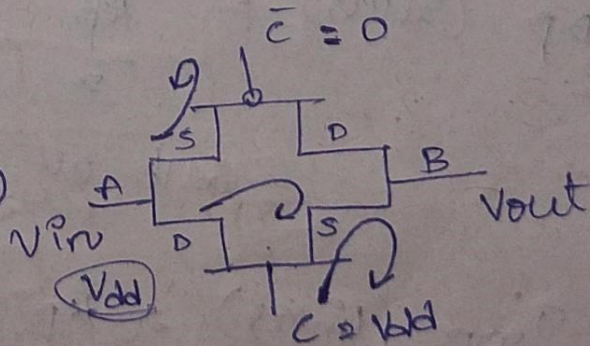
$A = V_{dd}, C = V_{dd}$

$V_{out} = B = V_{dd}$

In NMOS,

KVL

$$V_{DS, N} = V_{dd} - V_{out} \quad \text{--- (1)}$$



~~V_{DS}~~

$$V_{GS, N} = V_{DD} - V_{out} \quad \text{--- (2)}$$

For pmos,

$$V_{p, sd} = V_{DD} - V_{out}$$

but actually we say V_{dsp} so

$$V_{dsp} = V_{out} - V_{dd} \quad (3)$$

$$\cancel{V_{gsp}} = -V_{dd} + V_{sgp} + 0 = 0$$

$$V_{sgp} = V_{dd}$$

$$\Rightarrow V_{gsp} = -V_{dd}$$

$$\Rightarrow V_{gsp} = -V_{dd} \quad (4)$$

$$I_D = I_{SDP} + I_{DSN} \quad (5)$$

$$R_{eqP} = \frac{V_{DD} - V_{out}}{I_{SDP}} \quad (6)$$

$\downarrow$
Requlent P

$$R_{eqN} = \frac{V_{DD} - V_{out}}{I_{DSN}} \quad (7)$$

$$I_{SDP} \text{ (linear)} = \frac{k_p}{2} (2(V_{gs} - V_{thp})V_{dsp} - V_{dsp}^2)$$

$$I_{SDP} \text{ (satur)} = \frac{k_p}{2} (V_{gs} - V_{thp})^2$$

sub all (1) - (4)

$$I_{SDP} \text{ (linear)} = \frac{k_p}{2} (2(-V_{dd} - V_{thp})(V_{out} - V_{dd}))$$

$$I_{SDP} \text{ (satu)} = \frac{k_p}{2} (-V_{dd} - V_{thp})^2$$

(linear) $I_{DSN} = \frac{k_N}{2} (2(V_{GSN} - V_{thN})V_{DSN} - V_{DSN}^2)$

(sat) $I_{DSN} = \frac{k_N}{2} (V_{GSN} - V_{thN})^2$

starting

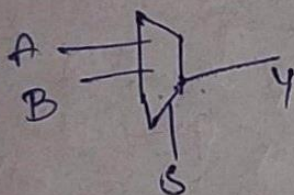
PMOS - sat
NMOS - sat

↓
PMOS - linear
NMOS - linear

↓
NMOS - cutoff

Delay = 0.69 R_{eq} C_L

2x1 MUX with Tg gate

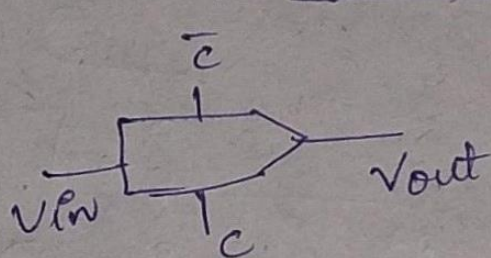


$Y = A\bar{C} + BC$

Truth table

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

$(V_{out} - V_{dd})^2$



↑
0

$S = 0$
 $Y = A$
op. nvali
ante
c=0

$S = 1$
 $Y = B$

2x1 MUX

